

PosTerminal-Centric Architecture Refactor

Branch: [cursor/full-postterminal-refactor-a435](#) / continued in package-layout
PR: <https://github.com/berothomas33/esaylinkrefactor/pull/3>

Goal

Make one PosTerminal API work with any card on any vendor.
UI and core never import Pax / Ingenico / Fake types.

End-state architecture

```
UI -> PosTerminal.acceptCard() / startTransaction()
    +-- EmvBehavior.prepare()
    +-- searchCard(config, CardSearchListener)  <- vendor readers
    +-- EmvBehavior.start(engine, config, card)  <- vendor EMV

EmvEngine - thin subjects + notify* -> behavior.dispatch*
```

Layer	Owns
`PosTerminal`	Device init, unified card search, cancel, host, printer
`EmvBehavior`	EMV after entry method selected
`EmvEngine`	Reactive events for UI (no business logic)

What changed

Unified card search

- Added EntryMethod, CardSearchListener, CardPresence
- Removed CardReader
- PosTerminal orchestrates: prepare -> searchCard(listener) -> behavior.start(card)
- Vendor terminals fire listener events from native SDKs

Card search listener events

- Search started
- Chip (ICC) detected
- Contactless (NFC/PICC) detected
- Magstripe detected
- Manual entry selected
- Card removed
- Search timeout / cancelled
- Reader error

Vendor layout

```
vendor/
  TerminalFactory.java
  pax/      PaxTerminal, PaxEmvBehavior, PaxKernel
  fake/     FakeTerminal, FakeEmvBehavior
```

```
ingenico/ IngenicoTerminal, IngenicoEmvBehavior (stub)
```

UI

- Primary action: Present card (any) -> PosTerminal.acceptCard() (Mode.ANY)
- Optional chip-only / contactless-only
- Cancel

Removed layers

- Per-vendor host/printer stub classes (later moved to PosTerminal ownership)
 - Router service lookup from app vendor façade
 - Contactless/Contact runners from app path
 - Callback adapters
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Adding a vendor

1. Implement XxxTerminal extends PosTerminal (searchCard, cancelCardSearch)
 2. Implement XxxEmvBehavior implements EmvBehavior
 3. One line in TerminalFactory
 4. No core / engine / UI changes
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Ingenico note

No Ingenico SDK was attached in the workspace. Stubs demonstrate the extension pattern; replace with Tetra/Axium when available. Wire with -PVENDOR=INGENICO.

Verification

:app:compileDebugJavaWithJavac succeeded for FAKE, INGENICO, and PAX.